Project Report

Web Application Vulnerability Scanning, Risk Evaluation & Alert System

1. Introduction

Modern web applications face numerous cybersecurity threats due to exposed services, misconfigurations, and vulnerable software components. Identifying these weaknesses early is critical to prevent attacks such as unauthorized access, data breaches, and service disruption.

This project implements a **Web Application Vulnerability Scanning and Risk Evaluation System** that scans target web hosts, analyzes exposed services, evaluates security risks, and visualizes results through an interactive dashboard. Additionally, an **automated email alert system** is integrated to notify users whenever high-risk vulnerabilities are detected.

The system integrates multiple tools such as **Nmap for network scanning, VirusTotal for threat intelligence, and Streamlit for dashboard visualization**, enabling users to easily understand the security posture of scanned targets.

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2. System Architecture

The system consists of three major components:

1. **Vulnerability Scanner**
2. **Risk Evaluation Dashboard**
3. **Automated Email Alert System**

The workflow of the system is as follows:

Target Hosts → Nmap Scan → Result Parsing → VirusTotal Intelligence → Risk Score Calculation → Dashboard Visualization → Email Alert (if High/Critical Risk)

The scanner probes selected test websites and identifies open ports and running services. These results are processed and analyzed to determine the risk level of each host.

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3. Design Decisions

3.1 Vulnerability Scanning

The scanning component was implemented using **Nmap**, a widely used network security scanner. Nmap performs service and port discovery on target hosts and exports results in **XML format**. XML output allows easy parsing using Python libraries.

Only **authorized test websites** such as testphp.vulnweb.com, testasp.vulnweb.com, and zero.webappsecurity.com were scanned to ensure ethical and legal compliance.

The scanner identifies security issues such as:

- Open insecure ports (FTP, Telnet)

- Exposed HTTP services
- Multiple exposed services on the same host
- Potentially risky services based on known attack surfaces

3.2 Threat Intelligence Integration

To enhance vulnerability analysis, the system integrates the **VirusTotal API**. VirusTotal provides reputation information about IP addresses and domains by aggregating threat intelligence from multiple security vendors.

The system queries VirusTotal for each discovered IP address and extracts the **number of malicious reports**. These reports are used to determine whether the host is associated with suspicious or malicious activity.

3.3 Risk Evaluation Model

A simple but effective **risk scoring model** was implemented to prioritize vulnerabilities.

Each detected service is assigned a risk score based on factors such as:

- Service type
- Known attack surface
- VirusTotal malicious reports

Example scoring:

- Telnet → High risk
- FTP → Medium risk
- HTTP/HTTPS → Moderate risk

The system calculates an overall **risk score per host** and assigns a severity level:

Risk Score	Severity
1–3	Low
4–6	Medium
7+	High

This helps users quickly identify critical systems that require immediate attention.

3.4 Interactive Dashboard

The dashboard was developed using **Streamlit**, which allows rapid development of interactive web applications using Python.

The dashboard includes:

- **Key security metrics** (total hosts, open ports, services, risk score)
- **Interactive charts** using Plotly
- **Threat intelligence panel**
- **Host summary table**
- **Filtering options**
- **Data export feature**

The dashboard is organized into navigation tabs:

- **Scan Data** – Raw scan results
- **Charts** – Visualization of risks and services
- **Threat Intelligence** – High-risk vulnerability alerts
- **Export** – Download scan results as CSV

To make the dashboard publicly accessible from Google Colab, **Cloudflare Tunnel** was used to expose the Streamlit server via a temporary public URL.

3.5 Email Alert System

An automated email alert system was implemented using Python's **smtplib** library.

Whenever a vulnerability with **High or Critical severity** is detected, the system automatically sends an HTML-formatted email containing:

- Target host
- Scan timestamp
- Risk score
- Table of high-risk vulnerabilities
- Recommended actions

This feature ensures that security issues are reported immediately without requiring manual monitoring.

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4. Challenges Faced

4.1 Integrating Multiple Tools

One challenge was integrating multiple tools such as **Nmap, VirusTotal API, and Streamlit** into a single pipeline. Each component produces different types of outputs, requiring careful data processing and formatting.

4.2 Handling API Limitations

VirusTotal API has **rate limits**, which restrict the number of requests that can be made within a specific time period. To avoid errors, the system had to be designed to handle API responses properly and manage request frequency.

4.3 Dashboard Design

Creating a professional cybersecurity-style dashboard required designing multiple visualizations and ensuring that the layout remained clean and easy to interpret. Streamlit's tab and sidebar components were used to organize the interface effectively.

4.4 Running the Dashboard from Google Colab

Google Colab does not directly support hosting web applications. Therefore, **Cloudflare Tunnel** was used to expose the Streamlit dashboard via a public URL. Configuring the tunnel and ensuring stable connectivity required additional setup.

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5. Conclusion

This project successfully demonstrates a **complete web vulnerability scanning and risk evaluation system**. By combining network scanning, threat intelligence, visualization, and automated alerts, the system provides an effective tool for understanding the security posture of web hosts.

The integration of Nmap, VirusTotal, Streamlit, and Cloudflare enables the system to perform automated scanning, analyze security risks, visualize findings, and notify users of critical vulnerabilities.

This approach highlights how cybersecurity tools can be combined into a **comprehensive security monitoring solution** that assists analysts in identifying and prioritizing potential threats.

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6. Tools and Technologies Used

- Python
- Nmap
- VirusTotal API
- Streamlit
- Plotly
- Pandas
- smtplib (Email alerts)
- Cloudflare Tunnel